

AMRUTHA KATTIMANI

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EDUCATION

PES University — RR Campus, Bangalore

2024 – 2028

- B.Tech in Computer Science & Engineering | CGPA: 8.125 / 10.0
- CNR Rao Merit Award — Top 20% of students | Merit Certificate & Cash Prize

EXPERIENCE

Research Intern — ISFCR SOC | PES University

2 Months

- Developed deep learning models to detect temporally manipulated sequences in video content for digital media forensics research, aimed at academic publication.
- Managed video dataset preprocessing, model training pipelines, and evaluation under research supervision at the Centre for Information Security, Forensics and Cyber Resilience.

PROJECTS

FirstIn — AI-Powered Placement Copilot

React, Node.js, Express, PostgreSQL, pgvector, Supabase, Groq Llama 3.1/3.3, HuggingFace, Playwright

- Gmail IMAP scraper extracts structured opportunity data via Groq Llama 3.1 8B with versioned prompts and A/B testing; full RAG pipeline using HuggingFace all-MiniLM-L6-v2 embeddings in pgvector powers Scout chatbot with semantic retrieval answered by Llama 3.3 70B.
- Collaborative filtering built from scratch (user-item matrix, cosine similarity) for personalised recommendations; Playwright agent auto-fills application forms; JWT auth, urgency-ranked feed, apply/skip tracking with undo history.

Replast AI — Plastic Circular Economy Platform

React, Node.js, Express, MongoDB, FastAPI, HuggingFace, Gemini API

- AI-powered marketplace connecting plastic waste contributors with buyers; FastAPI microservice classifies plastic type (PET, HDPE, PVC, LDPE, PP, PS) from uploaded images using a pretrained HuggingFace model.
- JWT auth with role-based access control, Cloudinary image storage, RESTful Express API on MongoDB Atlas, and multi-state procurement workflow (pending → accepted → dispatched → delivered).

DeepGuard AI — Multi-Agent Deepfake Video Forensics

Google ADK, Gemini, HuggingFace, FastAPI, Docker

- Built a Google ADK multi-agent system with specialised agents for video inspection, evidence analysis, and report generation using reusable Agent Skills and MCP tools.
- Designed a forensic investigation workflow with metadata analysis, structured report generation, and deployment-ready architecture using FastAPI and Docker.

TECHNICAL SKILLS

Languages	Python, C
Frontend	React, Tailwind CSS, Axios, React Router
Backend	Node.js, Express, FastAPI, JWT, REST APIs, Socket.IO
Databases	PostgreSQL, pgvector, Supabase, MongoDB Atlas
AI / ML	LangChain, HuggingFace, Groq API, Gemini API, Google ADK, OpenCV, PyTorch (basics)
Concepts	RAG, AI Agents, Collaborative Filtering, NLP Extraction, Computer Vision
Tools	Git, GitHub, Playwright, Docker, Cloudinary, Vercel, Render, Postman

ADDITIONALS

- DeepLearning.AI × OpenAI — ChatGPT Prompt Engineering for Developers.
- Google × Kaggle 5-Day AI Agents Intensive Hackathon (2026) — built a production-ready multi-agent agentic AI system as capstone.
- Other projects: Concurrent Seat Booking System, Mini Docker Container, Knowledge Builder, Leaf Health Detection, Smart Energy Monitor.
- Passionate about building scalable AI systems using RAG, Multi-Agent Architectures, and Agentic Workflows.